

$$[e = 1.6 \times 10^{-19}] \text{ C (Coulomb)} \quad \text{8-2 unit of charge}$$

Coulomb's Law

if two point charges q_1 & q_2 are separated by a distance r in vacuum, the magnitude of the force (F) b/w them.

$$F = k \frac{q_1 q_2}{r^2}$$

$$k = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$$

$$k = \frac{1}{4\pi\epsilon_0}$$

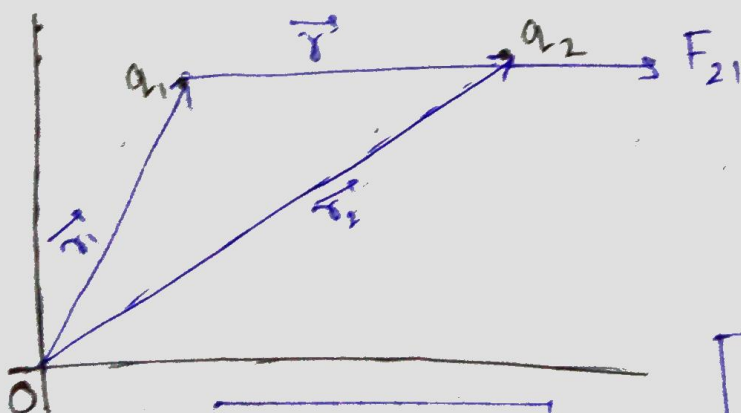
So

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

ϵ_0 permittivity of free space.

$$\epsilon_0 = 8.85 \text{ C}^2/\text{Nm}^2$$

In vector form [*]



$$\vec{r}_1 + \vec{r} = \vec{r}_2$$

$$\vec{r} = \vec{r}_2 - \vec{r}_1$$

$$|\vec{r}| = |\vec{r}_2 - \vec{r}_1|$$

$$|\vec{F}| = |\vec{F}| \hat{r}$$

$$\vec{F}_{21} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{|\vec{r}_{21}|^3} \vec{r}_{21}$$

in general

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{|\vec{r}|^3} \vec{r}$$